LEARNING MODULE

# 9

# Linear Systems, Inverses and Rank

*One system read four ways, by rows, columns, determinants and elimination, and why an exact fit on three days proves nothing.*

**VOLUME** II · Linear Algebra for Portfolios

**BUILDS ON** LM2 · LM6 · LM7 · LM8

**LEADS TO** LM10 · LM11 · LM12 · LM23 · LM27 · LM28 · LM43

**TIME** ≈ 2.5 h + lab

## LEARNING OUTCOMES

| *Mastery* | *After this module you should be able to:* |
|---|---|
| ☐ | 1 write a linear system $M\mathbf{x} = \mathbf{b}$ by rows and by columns, and read its solutions from the row picture and the column picture |
| ☐ | 2 calculate $2 \times 2$ and $3 \times 3$ determinants by cofactor expansion and by row operations, and interpret a determinant as a signed area or volume |
| ☐ | 3 solve small systems with Cramer's rule, and compare its cost with that of elimination |
| ☐ | 4 solve a system by Gaussian elimination with back substitution and row swaps, and compute its determinant from the pivots |
| ☐ | 5 compute an inverse with the $2 \times 2$ formula and by Gauss–Jordan elimination, and apply the product and transpose rules |
| ☐ | 6 determine the rank, column space and null space of a matrix, and classify a system as having no, one or infinitely many solutions |
| ☐ | 7 identify perfect collinearity in a matrix of returns, and explain why a daily-rebalanced mix creates it and a buy-and-hold mix does not |
| ☐ | 8 solve and test linear systems in NumPy with `solve`, `inv`, `det` and `matrix_rank`, and interpret their floating-point output |

**LEARNING MODULE OVERVIEW**

- A linear system $M\mathbf{x} = \mathbf{b}$ is a set of lines or planes that must meet (rows) and a combination of columns that must reach $\mathbf{b}$ (columns). On 8 and 9 April 2025, 1.035 of the Semiconductor index and 0.207 of the FinancialInsurance index reproduce TAIEX's returns on both days.
- A $2 \times 2$ determinant is the signed area spanned by the two columns: 24.753 for columns 47.15° apart, and 2.7813 for a pair only 1.52° apart, whose system needs multiples of −3.48 and 5.12.
- Cramer's rule, $x_j = \det M_j(\mathbf{b}) / \det M$, follows from the linearity of the determinant in each column. With cofactor expansions it needs $(n+1)!$ terms, each a product of $n$ entries, against about $\frac{2}{3}n^3$ operations for elimination: about $5.1 \times 10^{19}$ terms against 5,333 operations at $n = 20$.
- Gaussian elimination reduces the 7–9 April system to triangular form. Its exact pivots, −9.95, 4.495940 and 1.600370 to six decimals, multiply to the determinant −71.591922. NumPy's `solve` runs the same elimination, with row swaps.
- The inverse exists exactly when $\det M \neq 0$, when $\operatorname{rank} M = n$ and when only $\mathbf{x} = \mathbf{0}$ solves $M\mathbf{x} = \mathbf{0}$. In code `solve` comes first; `inv` serves concepts such as LM7's Kelly fractions $\Sigma^{-1}(\boldsymbol{\mu} - r_f\mathbf{1}) = (5.59,\ -0.52)^{\top}$.
- A system has a solution exactly when $\operatorname{rank} M = \operatorname{rank}[M \mid \mathbf{b}]$, and its solutions are then $\mathbf{x}_p +$ null $M$. With three sectors, two days give a line of mixes, three days exactly one and four days none.
- Any three days with a non-zero determinant fit exactly, so an exact fit carries no evidence. FinancialInsurance's multiple runs from −1.479 to 2.069 across the four triples of days, and the 7–9 April mix gives +10.17% for 10 April, against +9.25%.
- A return matrix has rank below its number of columns exactly when a fixed non-zero combination of the columns is 0 on every date. A daily-rebalanced 50/50 mix of Semiconductor and FinancialInsurance creates one (rank 2); a buy-and-hold mix, whose Semiconductor weight drifts from 0.50 to 0.858, does not (rank 3).

# 1 INTRODUCTION

A replicating mix asks which multiples of a few assets reproduce a target's returns. Each day gives one linear equation in the multiples, so a few days give a linear system. This module solves such systems four ways and builds the inverse and the rank from them. It also shows why an exact fit on a few days says almost nothing.

**MOTIVATING CASE** **Three sectors, three days**

On 7 April 2025, after the 3–4 April holidays, TAIEX fell 9.70%, its largest daily move in data back to July 2007. It then moved −4.02%, −5.79% and +9.25% on 8, 9 and 10 April. Is there one mix of the Semiconductor, FinancialInsurance and Food sector indices whose returns reproduce TAIEX's on each day (Exhibit 1)?

With two days there are infinitely many such mixes. With 7–9 April there is exactly one: 0.334 of Semiconductor, 1.447 of FinancialInsurance and −0.916 of Food, which sum to 0.865. With 8–10 April the mix becomes $(-0.018,\ 2.069,\ -1.376)^{\top}$. With all four days there is none, and the 7–9 April mix gives +10.17% on 10 April, not +9.25%. Why does a perfect fit say so little?

**Exhibit 1: TAIEX and three sector indices, 7–10 April 2025: daily returns in percent**

| Session | From the close of | TAIEX $b_i$ | Semiconductor $m_{i1}$ | FinancialInsurance $m_{i2}$ | Food $m_{i3}$ |
|---|---|---|---|---|---|
| 7 April | 2 April | −9.70 | −9.95 | −9.89 | −8.66 |
| 8 April | 7 April | −4.02 | −3.99 | 0.53 | 3.77 |
| 9 April | 8 April | −5.79 | −4.47 | −5.61 | −4.17 |
| 10 April | 9 April | 9.25 | 9.91 | 8.36 | 5.72 |

Source: FinMind `TaiwanStockPrice`, closes of TAIEX and of the TWSE sector price indices, retrieved 11 October 2026; no close in this window is repaired. Returns are $100(P_t/P_{t-1}-1)$ between consecutive trading dates, rounded to 0.01; these rounded values are the inputs of every system in this module. The unrounded return closest to a rounding tie, Food on 8 April at 3.774862, lies 0.000138 from it. Computations: `data/lm09_numbers.py`.

Sections 2–6 draw the two pictures of a system, build the determinant, Cramer's rule, elimination and the inverse, and solve the case. Section 7 counts solutions with the rank, and Section 8 turns rank into perfect collinearity of returns.

**NOTE** **Notation for this module**

A system $M\mathbf{x} = \mathbf{b}$ has $m$ equations and $n$ unknowns; $M$ is $m \times n$, with entries $m_{ij}$ (row $i$, column $j$) and columns $\mathbf{m}_j$, and $[M \mid \mathbf{b}]$ is the augmented matrix. The unknown $\mathbf{x} = (x_1, x_2, x_3)^\top$ of the case is the **replicating mix** of Semiconductor, FinancialInsurance and Food. Its multiples need not sum to 1, so they are not portfolio weights.

$M_j(\mathbf{b})$ is $M$ with column $j$ replaced by $\mathbf{b}$, $M^{(ij)}$ is $M$ without row $i$ and column $j$, adj $M$ is the adjugate and $\mathbf{k}_j$ is column $j$ of $M^{-1}$. col $M$, null $M$ and dim are the column space, the null space and a dimension; $\mathbf{x}_p$ is a particular solution and $\mathbf{x}_h$ a null-space vector whose free variable is 1. $\ell_{ik}$ is a multiplier and $d_k$ the pivot of elimination step $k$; $\theta$ is an angle and $a$ a constant. In Problem 8, $v_t = (1 + z_t)^{-t}$ is the discount factor for $t$ years at LM6's annual-compounding spot rate $z_t$ (LM6's symbol; LM12's $\mathbf{z}_t$ is a whitened return vector).

Example 6 and the remarks on it use LM7's covariance matrix $\Sigma$ (entries $\sigma_1^2$, $\sigma_{12}$, $\sigma_2^2$), expected returns $\boldsymbol{\mu}$, rate $r_f$ and Kelly fractions $f_1^*$, $f_2^*$. Only Section 8 uses the $T \times N$ return matrix $X$ of LM8, returns $R_{S,t}$, $R_{F,t}$ of Semiconductor and FinancialInsurance on date $t$, and a weight $w_t$. Counts: $m$, $n$; indices $i$, $j$, $k$; the time index $t$ counts dates in Section 8 and years in Problem 8.

## 2 TWO PICTURES OF A LINEAR SYSTEM

- [ ] write a linear system $M\mathbf{x} = \mathbf{b}$ by rows and by columns, and read its solutions from the row picture and the column picture

Start with two days and two unknowns. Which multiple $x_1$ of Semiconductor and $x_2$ of FinancialInsurance reproduce TAIEX's returns on 8 and 9 April? Each day gives one equation:

$$\begin{aligned} -3.99\,x_1 + 0.53\,x_2 &= -4.02 \quad (8\text{ April}),\\ -4.47\,x_1 - 5.61\,x_2 &= -5.79 \quad (9\text{ April}). \end{aligned}$$

**DEFINITION** **Linear system**

A **linear system** of $m$ equations in $n$ unknowns is $M\mathbf{x} = \mathbf{b}$, with $M$ an $m \times n$ matrix, $\mathbf{x}$ in $\mathbb{R}^n$ and $\mathbf{b}$ in $\mathbb{R}^m$. Row $i$ is the equation $\sum_j m_{ij}x_j = b_i$. The **augmented matrix** $[M \mid \mathbf{b}]$ appends $\mathbf{b}$ to $M$ as a last column.

**The row picture.** With two unknowns, each equation is a line in the $(x_1, x_2)$ plane, and a solution lies on every line at once. Here the two lines meet in one point (Exhibit 2). With three unknowns each equation is a plane, and three planes usually meet in one point.

**The column picture.** Read $M\mathbf{x}$ by columns instead, as LM8 (8.12) read a return matrix times a weight vector:

$$M\mathbf{x} = x_1\mathbf{m}_1 + x_2\mathbf{m}_2 + \cdots + x_n\mathbf{m}_n. \tag{9.1}$$

The system asks which combination of the columns reaches $\mathbf{b}$. In the case, column $j$ holds sector $j$'s returns on the chosen days. So a solution is a mix of sectors whose returns equal TAIEX's day by day (Exhibit 3).

**EXAMPLE 1**

## Two days, two sectors, two pictures

Solve the 8–9 April system above, and check the answer in both pictures.

**SOLUTION**

Eliminate one unknown at a time. Adding 5.61 times the first equation to 0.53 times the second removes $x_2$, with

$$\begin{aligned} -3.99(5.61) - 4.47(0.53) &= -24.753, \\ -4.02(5.61) - 5.79(0.53) &= -25.6209, \end{aligned}$$

so $-24.753\,x_1 = -25.6209$ and $x_1 = 1.035062$. Adding 4.47 times the first equation to −3.99 times the second removes $x_1$ and gives $24.753\,x_2 = 5.1327$, so $x_2 = 0.207357$. Both divisions use 24.753, which Section 3 names the determinant.

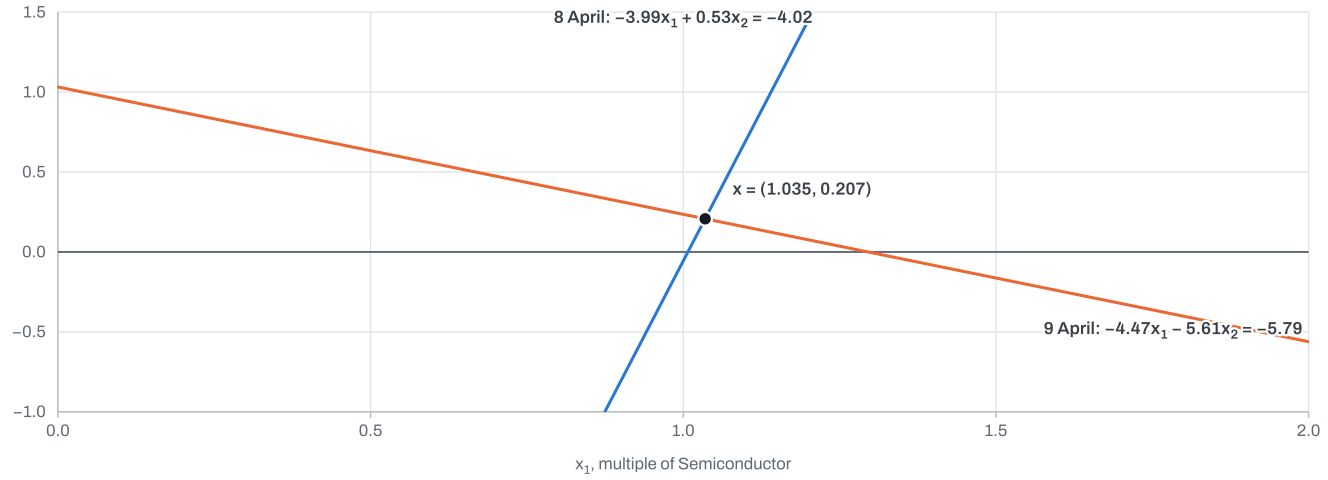
In the row picture the two lines cross at $(1.035062,\ 0.207357)$. In the column picture,

$$\begin{aligned} 1.035062\,\mathbf{m}_1 + 0.207357\,\mathbf{m}_2 &= \begin{pmatrix} -4.02 \\ -5.79 \end{pmatrix} \\ &= \mathbf{b}. \end{aligned}$$

So 1.035 of Semiconductor and 0.207 of FinancialInsurance reproduce TAIEX's returns on both days.

## Exhibit 2: Row picture of the 8–9 April system: two lines meet in one point

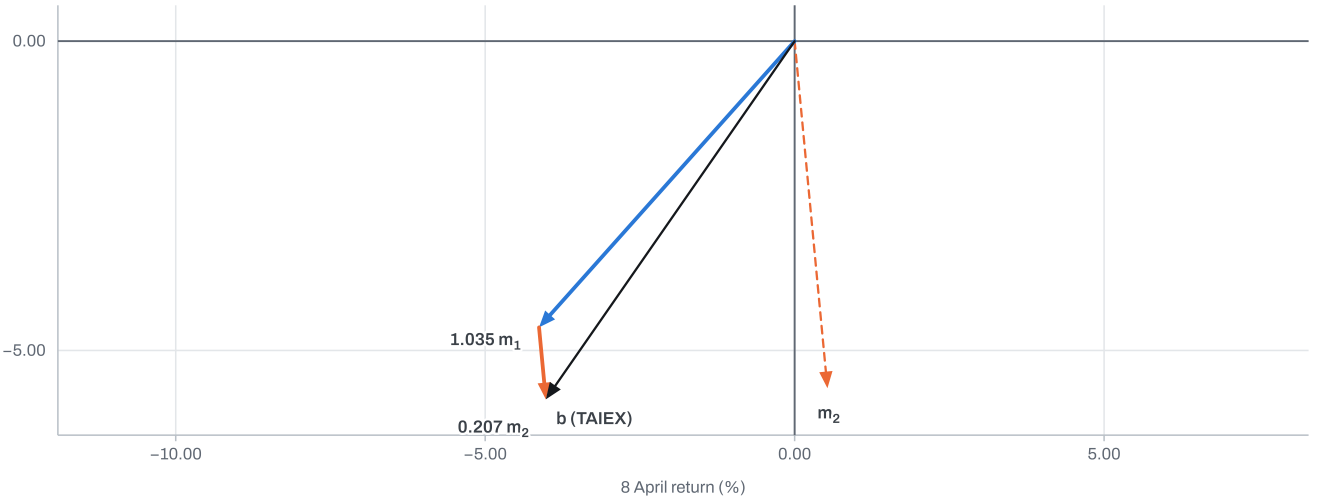
$x_2$, multiple of FinancialInsurance




Note: each line holds the mixes $(x_1, x_2)$ of Semiconductor and FinancialInsurance that reproduce TAIEX on one day. The solution of Example 1 lies on both.

## Exhibit 3: Column picture of the same system: scaled columns that add up to $\mathbf{b}$

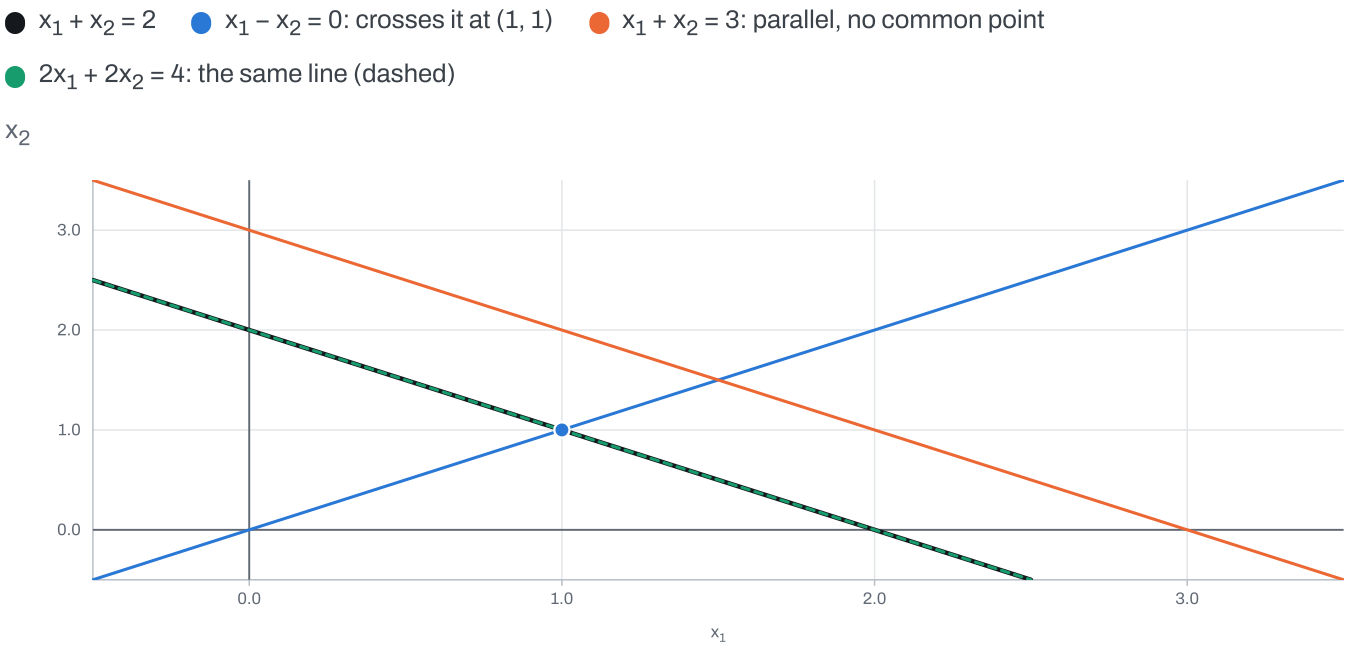
9 April return (%)




Note: $\mathbf{m}_1$ holds Semiconductor's and $\mathbf{m}_2$ FinancialInsurance's returns on 8 and 9 April, in percent. The arrow $1.035\,\mathbf{m}_1$ plus the arrow $0.207\,\mathbf{m}_2$, drawn from its tip, ends at TAIEX's $\mathbf{b}$.

Two lines in a plane can meet in one point, be parallel or coincide. So a $2 \times 2$ system has exactly one solution, none or infinitely many (Exhibit 4). Section 7 proves that the same three cases cover every linear system.

**Exhibit 4: Three partners for the line $x_1 + x_2 = 2$**




Note: illustrative equations. A crossing partner gives one solution and a parallel partner none. The coincident partner, the same equation times 2 (dashed), gives every point of the line.

**導讀** **兩種讀法：一列一個方程，一行一個產業**

一個線性方程組（linear system）可以按列讀，也可以按行讀。按列讀，每一天是一條直線（三個未知數時是一個平面），解就是所有直線的交點。按行讀，每一行是一個產業指數在這幾天的報酬，解就是用這幾個產業「組合」出 TAIEX 報酬的倍數，也就是一個線性組合（linear combination）。兩條直線可能交於一點、平行或重合，所以解可能剛好一個、沒有或無限多個。

**KNOWLEDGE CHECK 1**

A $2 \times 2$ system has no solution. What do its two lines look like in the row picture, and where does $\mathbf{b}$ lie in the column picture?

**ANSWER**

The lines are parallel and distinct, so no point lies on both. In the column picture both columns lie on one line through the origin, and $\mathbf{b}$ lies off it, so no combination reaches $\mathbf{b}$.

## 3 THE DETERMINANT

- [ ] calculate $2 \times 2$ and $3 \times 3$ determinants by cofactor expansion and by row operations, and interpret a determinant as a signed area or volume

The number 24.753 that both divisions of Example 1 used is $m_{11}m_{22} - m_{12}m_{21}$. It is the signed area that the two columns span. It is 0 exactly when they lie on one line, but its size also depends on their lengths (see the pitfall below).

**DEFINITION** **Determinant of a $2 \times 2$ matrix**

For a $2 \times 2$ matrix $M$, the **determinant** is

$$\det M := m_{11}m_{22} - m_{12}m_{21}. \tag{9.2}$$

The absolute value of $\det M$ is the area $\|\mathbf{m}_1\| \, \|\mathbf{m}_2\| \sin\theta$ of the parallelogram spanned by the columns. The sign is positive when $\mathbf{m}_2$ lies counter-clockwise from $\mathbf{m}_1$.

The area follows from LM8's dot product. Expanding both sides shows

$$(\det M)^2 + (\mathbf{m}_1^{\top}\mathbf{m}_2)^2 = \|\mathbf{m}_1\|^2 \, \|\mathbf{m}_2\|^2,$$

and $\mathbf{m}_1^{\top}\mathbf{m}_2 = \|\mathbf{m}_1\| \, \|\mathbf{m}_2\| \cos\theta$, so $|\det M| = \|\mathbf{m}_1\| \, \|\mathbf{m}_2\| \sin\theta$. The determinant is 0 exactly when the columns lie on one line: $\theta$ is $0^\circ$ or $180^\circ$, or a column is $\mathbf{0}$.

**EXAMPLE 2**

## Area and nearly parallel columns

Find the determinant of Example 1 and the angle between its columns. Then solve the system of 7 and 9 April with Semiconductor and Food, and compare.

$$\begin{pmatrix} -9.95 & -8.66 \\ -4.47 & -4.17 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -9.70 \\ -5.79 \end{pmatrix}$$

**SOLUTION**

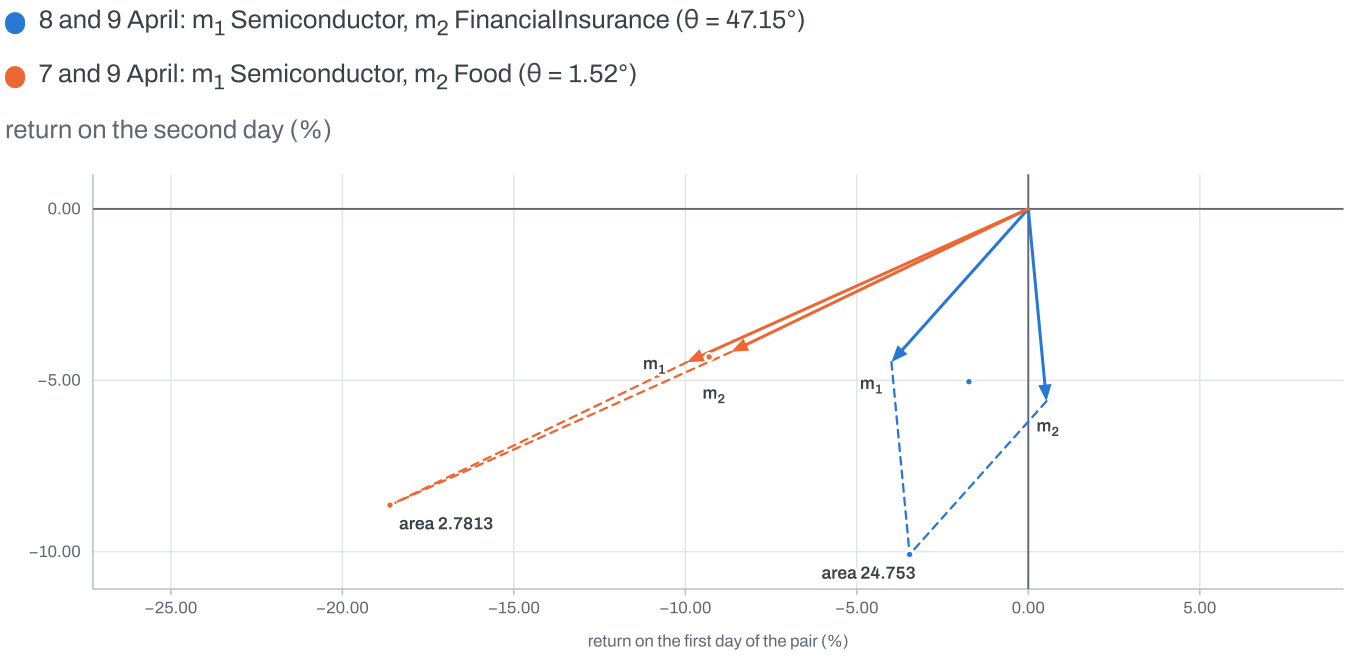
For Example 1,

$$\begin{aligned}\det M &= (-3.99)(-5.61) \\ &\quad - (0.53)(-4.47) \\ &= 22.3839 + 2.3691 = 24.753.\end{aligned}$$

The columns have lengths 5.991744 and 5.634980, so $\sin\theta = 24.753/33.7634 = 0.7331$. The dot product of the columns, 22.962, is positive, so $\theta$ is acute and equals 47.15°.

For Semiconductor and Food, $\det M = 41.4915 - 38.7102 = 2.7813$. The columns are longer, 10.907951 and 9.611686, but $\sin\theta = 2.7813/104.8438 = 0.0265$. Their dot product, 104.8069, is also positive, so they lie only 1.52° apart (Exhibit 5). Eliminating as in Example 1 gives $x_1 = -9.6924/2.7813 = -3.48$ and $x_2 = 14.2515/2.7813 = 5.12$. To reproduce TAIEX with two nearly parallel columns, the mix shorts 3.48 of Semiconductor against 5.12 of Food.

**Exhibit 5: Two parallelograms: columns 47.15° apart and columns 1.52° apart**




Note: each arrow is one sector's returns on two days, in percent; the dashed sides complete each parallelogram, whose area is $|\det M|$. The thin one has the longer columns but about a ninth of the area.

For larger matrices the **cofactor expansion** defines the determinant step by step, along the first row:

$$\det M = \sum_{j=1}^{n} (-1)^{1+j}\, m_{1j}\, \det M^{(1j)}.$$

Each $(-1)^{i+j} \det M^{(ij)}$ is a **cofactor**. Expanding along any other row or column gives the same value. For $n = 3$ the absolute value is the volume of the box spanned by the columns (both stated without proof;

Strang 2016).

**THEOREM** **Row operations and the determinant**

Let $M$ and $K$ be $n \times n$ matrices.

1. Swapping two rows changes the sign of the determinant.
2. Multiplying one row by a number $a$ multiplies the determinant by $a$.
3. Adding a multiple of one row to another leaves the determinant unchanged.
4. The determinant of a triangular matrix is the product of its diagonal entries.

Moreover,

$$\begin{aligned} \det(MK) &= \det M \, \det K, \\ \det M^{\top} &= \det M, \\ \det M^{-1} &= 1/\det M, \end{aligned} \tag{9.3}$$

the last when $M^{-1}$ exists. Rules 1–4 are stated without proof (Strang 2016); the Deep Dive below derives (9.3) from them.

Since $\det M^{\top} = \det M$, the rules hold for columns too. If one column is a combination of the others, the columns are **dependent** (Section 7 defines independence). Subtracting that combination (rule 3) then leaves a zero column, so $\det M = 0$. A square matrix with $\det M = 0$ is **singular**; Section 5 shows that its columns are dependent.

**EXAMPLE 3**

## The determinant of the 7–9 April system

With rows 7, 8 and 9 April and columns Semiconductor, FinancialInsurance and Food, the augmented matrix $[M \mid \mathbf{b}]$ of the case is

$$\left(\begin{array}{ccc|c} -9.95 & -9.89 & -8.66 & -9.70 \\ -3.99 & 0.53 & 3.77 & -4.02 \\ -4.47 & -5.61 & -4.17 & -5.79 \end{array}\right).$$

Compute $\det M$ by cofactor expansion along the first row.

**SOLUTION**

The three minors are

$$\begin{aligned}\det M^{(11)} &= 0.53(-4.17) - 3.77(-5.61)\\ &= 18.9396,\\ \det M^{(12)} &= 3.99(4.17) + 3.77(4.47)\\ &= 33.4902,\end{aligned}$$

and $\det M^{(13)} = 24.753$, the determinant of Example 1. So

$$\begin{aligned}\det M &= -9.95(18.9396)\\ &\quad + 9.89(33.4902) - 8.66(24.753)\\ &= -71.591922,\end{aligned}$$

the sum of −188.44902, 331.218078 and −214.36098. The determinant is not 0, so the three columns are independent, and Section 5 finds exactly one mix.

**PITFALL** **A small determinant does not mean nearly dependent columns**

The determinant scales with the entries: $\det(0.1\, I_9) = 10^{-9}$, yet the system $0.1\,\mathbf{x} = \mathbf{b}$ has the plain solution $\mathbf{x} = 10\,\mathbf{b}$. Judge a determinant against the column lengths instead: 24.753 against 33.76 in Example 2 is far from 0, and 2.7813 against 104.84 is close. LM11 measures this closeness with the condition number, which tells how strongly a nearly singular matrix magnifies small errors.

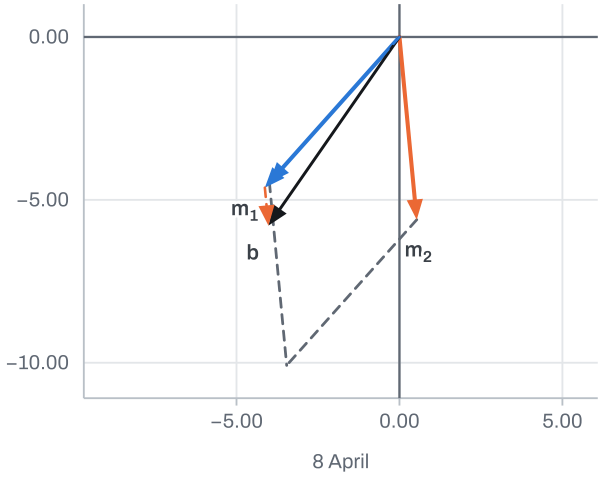
In the explorer below, turn the second column toward the first and watch the area shrink and the mix grow.

## A 2×2 system in two pictures: columns, rows and the determinant

INTERACTIVE

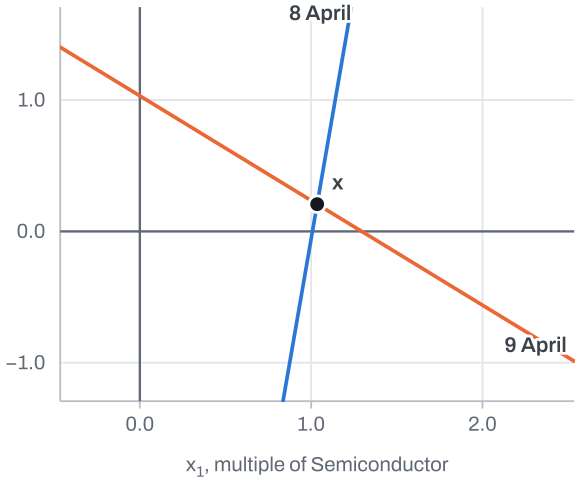
Column picture

9 April




Row picture

$x_2$, multiple of FinancialInsurance




| det M = $m_{11}m_{22} - m_{12}m_{21}$ | Angle between the columns | rank M |
|---|---|---|
| **24.753** | **47.15°** | **2** |
| **Solutions** | **$x_1$ (Semiconductor)** | **$x_2$ (FinancialInsurance)** |
| **exactly one solution** | **1.0351** | **0.20736** |

Column picture: the columns $m_1$ (fixed by the system chosen) and $m_2$ (sliders) span a parallelogram whose area is |det M|; when a solution exists, the dashed path $x_1m_1 + x_2m_2$ ends at b. Row picture: one line per equation, meeting at the solution. As $m_2$ turns toward $m_1$ the area shrinks and the mix grows: LM11 measures this sensitivity with the condition number. The readout rank M is the number of independent columns (Section 7). A determinant within $10^{-9}$ of 0, relative to the product of the column lengths, counts as 0.

### KNOWLEDGE CHECK 2

Swap the rows of 7 and 8 April in Example 3, with their entries of **b**. What is the new determinant, and does the mix change?

**ANSWER**

By rule 1 the determinant becomes +71.591922. The mix does not change, because the swap only reorders the equations.

**DEEP DIVE** **The product rule (9.3)** optional · click to expand

A row operation on $K$ is a product $EK$, where the **elementary matrix** $E$ is $I_n$ with the same operation applied. By rules 1–3, $\det(EK) = \det E \, \det K$. If $M$ is invertible, Gauss–Jordan elimination (Section 6) writes it as a product $E_1 E_2 \cdots$ of elementary matrices. Applying the rule once per factor gives $\det(MK) = \det E_1 \det E_2 \cdots \det K = \det M \det K$.

If $M$ is singular, so is $MK$, whose columns combine those of $M$; both sides are 0. For invertible $M$, transposing gives $M^\top = \cdots E_2^\top E_1^\top$, and transposing an elementary matrix keeps its determinant, so $\det M^\top = \det M$. If $M$ is singular, its columns are dependent (Section 5). They are the rows of $M^\top$, so rule 3 turns one of these rows into a zero row, and $\det M^\top = 0 = \det M$. Finally, $\det M \det M^{-1} = \det I_n = 1$ (Strang 2016).

## 4 CRAMER'S RULE

- [ ] solve small systems with Cramer's rule, and compare its cost with that of elimination

With the other columns held fixed, the determinant is linear in each column, as the cofactor expansion along that column shows. It is 0 when two columns are equal, because swapping them changes the sign but not the matrix. Now suppose $M\mathbf{x} = \mathbf{b}$, and replace column $j$ of $M$ by $\mathbf{b} = \sum_k x_k \mathbf{m}_k$. By linearity in column $j$,

$$\det M_j(\mathbf{b}) = \sum_k x_k \det M_j(\mathbf{m}_k) = x_j \det M,$$

because for $k \neq j$ the matrix $M_j(\mathbf{m}_k)$ holds column $\mathbf{m}_k$ twice.

**THEOREM** **Cramer's rule**

If $M$ is $n \times n$ and $\det M \neq 0$, the system $M\mathbf{x} = \mathbf{b}$ has exactly one solution, with entries

$$x_j = \frac{\det M_j(\mathbf{b})}{\det M}, \qquad j = 1, \ldots, n. \tag{9.4}$$

The derivation above shows that no other solution exists; the Deep Dive of Section 6 shows that (9.4) solves the system.

**EXAMPLE 4**

### Cramer's rule for the 7–9 April mix

Solve the system of Example 3 by (9.4).

**SOLUTION**

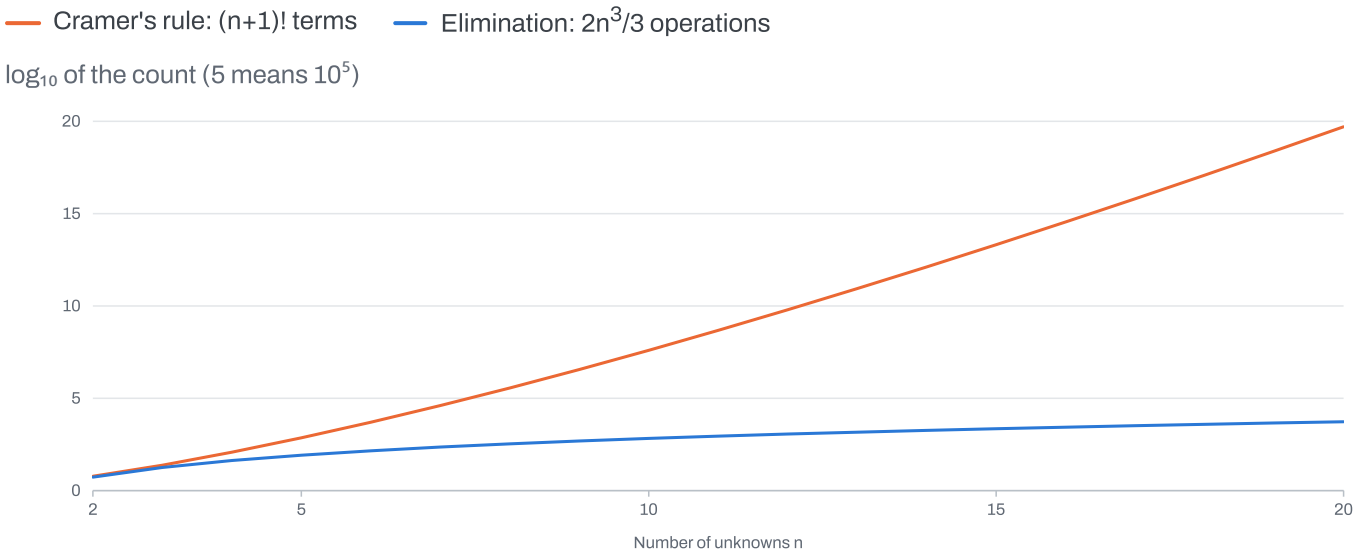
Replacing each column of $M$ in turn by $\mathbf{b}$ and expanding as in Example 3 gives $\det M_1(\mathbf{b}) = -23.919201$, $\det M_2(\mathbf{b}) = -103.581657$ and $\det M_3(\mathbf{b}) = 65.586258$. Dividing each by $\det M = -71.591922$,

$$\mathbf{x} = (0.334105,\ 1.446834,\ -0.916113)^{\top}.$$

The mix holds 0.334 of Semiconductor, 1.447 of FinancialInsurance and −0.916 of Food, and its multiples sum to 0.865, not 1. LM7 (7.12) is the same rule for two Kelly fractions. Each numerator there is the determinant of the covariance matrix with one column replaced by the excess returns.

Cramer's rule is exact and short to write, but slow to compute. A cofactor expansion of an $n \times n$ determinant has $n!$ terms, each a product of $n$ entries, and (9.4) needs $n+1$ determinants: $(n+1)!$ terms. Each term alone takes $n-1$ multiplications, so the number of terms understates the work. Elimination needs about $\frac{2}{3}n^3$ operations (Section 5 counts them). At $n = 20$ the term count alone is about $5.1 \times 10^{19}$, against about 5,333 operations for elimination (Exhibit 6).

**Exhibit 6: The work of Cramer's rule and of elimination for $n = 2, \ldots, 20$ unknowns**




Note: the vertical axis shows $\log_{10}$ of each count. Cramer's rule takes $(n+1)!$ terms, each a product of $n$ entries, and each term takes $n-1$ multiplications, so its curve understates its work. Elimination takes about $\frac{2}{3}n^3$ operations. At $n = 20$ the counts are $21! = 51{,}090{,}942{,}171{,}709{,}440{,}000$ and about 5,333.

**導讀** **行列式與克拉瑪法則**

$2 \times 2$ 行列式（determinant）就是兩個行向量（column vector）張成的平行四邊形（parallelogram）面積，正負號表示方向；$3 \times 3$ 時是平行六面體（parallelepiped）的體積。行列式為 0，表示某一行是其他行的線性組合（linear combination），方程組不會剛好只有一個解。克拉瑪法則（Cramer's rule）把每個未知數寫成兩個行列式的比，公式漂亮，二元、三元時手算很方便。但 $n$ 個未知數要算 $n+1$ 個行列式、共 $(n+1)!$ 項，$n = 20$ 時約 $5 \times 10^{19}$ 項，消去法（elimination）只要約五千次運算。行列式很小也不代表接近奇異（singular），要和行向量的長度比較，LM11 用條件數（condition number）衡量。

**KNOWLEDGE CHECK 3**

How many products of four entries does (9.4) need for four unknowns when every determinant is a cofactor expansion?

**ANSWER**

Five determinants of $4! = 24$ terms each make 120 products. Formed one by one at three multiplications each, they take 360 multiplications. A nested cofactor expansion reuses partial products and needs 40 multiplications per determinant, 200 in all.

## 5 GAUSSIAN ELIMINATION

- ☐ solve a system by Gaussian elimination with back substitution and row swaps, and compute its determinant from the pivots

Three operations on the equations leave the solutions of a system unchanged, because each can be undone. They swap two equations, multiply one by a non-zero number, or add a multiple of one equation to another. **Gaussian elimination** uses them to make the matrix upper triangular, with zeros below the diagonal. **Back substitution** then solves the last equation first and works upward.

At step $k$ the **pivot** $d_k$ is the diagonal entry of row $k$. For each row $i$ below it, the **multiplier** $\ell_{ik}$ is that row's entry in column $k$ divided by $d_k$. Subtracting $\ell_{ik}$ times row $k$ from row $i$ puts a zero there.

Step $k$ changes the $(n-k)^2$ entries below and to the right of the pivot, each with one multiplication and one subtraction. Summing over the steps,

$$\sum_{k=1}^{n-1} 2(n-k)^2 = \tfrac{2}{3}n^3 - n^2 + \tfrac{1}{3}n \approx \tfrac{2}{3}n^3$$

operations, with an error of order $n^2$; the right-hand side and back substitution add terms of the same order.

If a pivot is 0, swap its row with a lower row that has a non-zero entry in that column. Library code goes further. At every step it brings into the pivot position the row whose entry in the pivot column is largest in absolute value. This **partial pivoting** keeps every multiplier between −1 and 1, which limits the growth of rounding errors (Golub and Van Loan 2013, §3.4).

Elimination also delivers the determinant. Subtracting multiples of rows leaves it unchanged (rule 3), each swap flips its sign (rule 1), and the final matrix is triangular (rule 4). So

$$\det M = \pm\, d_1 d_2 \cdots d_n,$$

with a minus sign when the number of swaps is odd. With independent columns, elimination always finds a non-zero pivot. Otherwise the current column, zero at and below the pivot row, would be a combination of the earlier columns. So $\det M \neq 0$ for independent columns, the converse promised in Section 3.

**EXAMPLE 5**

## Elimination on the 7–9 April system

Solve the system of Example 3 by elimination, and check its determinant.

**SOLUTION**

The first pivot is $d_1 = -9.95$. The multipliers $\ell_{21} = -3.99/(-9.95) = 0.401005$ and $\ell_{31} = -4.47/(-9.95) = 0.449246$ clear column 1. Then the exact multiplier $\ell_{32} = -0.259557$ clears column 2 (Exhibit 7). Dividing the rounded entry −1.166955 by the rounded pivot 4.495940 gives −0.259558 instead. No swap is needed, and partial pivoting would choose the same rows. Back substitution in exact arithmetic gives

$$\begin{aligned} x_3 &= \frac{-1.466119}{1.600370} = -0.916113, \\ x_2 &= \frac{-0.130251 - 7.242704\, x_3}{4.495940} \\ &= 1.446834, \\ x_1 &= \frac{-9.70 + 9.89\, x_2 + 8.66\, x_3}{-9.95} \\ &= 0.334105, \end{aligned}$$

as in Example 4. With the rounded entries and the printed $x_3$ and $x_2$, the same steps give $x_2 = 1.446835$ and $x_1 = 0.334106$. The exact pivots multiply to $-71.591922 = \det M$.

**Exhibit 7: Elimination on $[M \mid \mathbf{b}]$ for 7–9 April**

| Step | Row | $x_1$ | $x_2$ | $x_3$ | $\mathbf{b}$ | Multiplier |
|---|---|---|---|---|---|---|
| 0 | 7 April | **−9.95** | −9.89 | −8.66 | −9.70 | |
| 0 | 8 April | −3.99 | 0.53 | 3.77 | −4.02 | |
| 0 | 9 April | −4.47 | −5.61 | −4.17 | −5.79 | |
| 1 | 8 April | 0 | **4.495940** | 7.242704 | −0.130251 | $\ell_{21} = 0.401005$ |
| 1 | 9 April | 0 | −1.166955 | −0.279528 | −1.432312 | $\ell_{31} = 0.449246$ |
| 2 | 9 April | 0 | 0 | **1.600370** | −1.466119 | $\ell_{32} = -0.259557$ |

Note: step $k$ subtracts $\ell_{ik}$ times row $k$ from row $i$; pivots in bold. Exact arithmetic on the inputs of Exhibit 1, rounded to six decimals; a step recomputed from the rounded entries can differ in the sixth decimal.

NumPy's `np.linalg.solve(M, b)` runs this algorithm, with partial pivoting. It calls LAPACK's `gesv`, which stores the multipliers and the triangular result as the LU factorization of $M$ and then substitutes (LAPACK, `dgesv`).

**KNOWLEDGE CHECK 4**

Eliminate $x_1$ from the second equation of Example 1. Give $\ell_{21}$, the new second row and $x_2$.

**ANSWER**

The multiplier is $\ell_{21} = -4.47/(-3.99) = 1.120301$, and the new second row of $[M \mid \mathbf{b}]$ is $(0,\ -6.203759 \mid -1.286391)$. So, as before,

$$\begin{aligned} x_2 &= \frac{-1.286391}{-6.203759} = 0.207357, \\ \det M &= (-3.99)(-6.203759) \\ &= 24.753. \end{aligned}$$

# 6 THE INVERSE

- [ ] compute an inverse with the $2 \times 2$ formula and by Gauss–Jordan elimination, and apply the product and transpose rules

**DEFINITION** **Inverse**

An $n \times n$ matrix $M$ is **invertible** if some $n \times n$ matrix $K$ satisfies $MK = KM = I_n$. That $K$ is unique and is written $M^{-1}$.

If $K$ and $K_2$ both qualify, then $K = K(MK_2) = (KM)K_2 = K_2$, so the inverse is unique. When it exists, the system $M\mathbf{x} = \mathbf{b}$ has the one solution $\mathbf{x} = M^{-1}\mathbf{b}$. For a $2 \times 2$ matrix with $\det M \neq 0$, multiplying out checks

$$M^{-1} = \frac{1}{\det M}\begin{pmatrix} m_{22} & -m_{12} \\ -m_{21} & m_{11} \end{pmatrix}. \tag{9.5}$$

For larger matrices, **Gauss–Jordan elimination** row-reduces $[M \mid I_n]$ until the left block is $I_n$; the right block is then $M^{-1}$. It solves the $n$ systems $M\mathbf{k}_j = \mathbf{e}_j$ at once, whose solutions $\mathbf{k}_j$ are the columns of $M^{-1}$. For invertible $M$ and $K$, two rules follow from the definition:

$$(MK)^{-1} = K^{-1}M^{-1},$$
$$(M^{\top})^{-1} = (M^{-1})^{\top}.$$

The first holds because $MK(K^{-1}M^{-1}) = I_n$ and $(K^{-1}M^{-1})MK = I_n$. The second follows from transposing $MM^{-1} = M^{-1}M = I_n$ with LM8 (8.15).

For an $n \times n$ matrix $M$ the following are equivalent:

$$\begin{aligned} M \text{ is invertible} &\iff \det M \neq 0 \\ &\iff \operatorname{rank} M = n \\ &\iff \operatorname{null} M = \{\mathbf{0}\}. \end{aligned} \tag{9.6}$$

Here the rank is the number of independent columns, and the null space is the set of solutions of $M\mathbf{x} = \mathbf{0}$ (Section 7). An inverse turns $M\mathbf{x} = \mathbf{0}$ into $\mathbf{x} = \mathbf{0}$. Independent columns give $n$ non-zero pivots and $\det M \neq 0$ (Section 5). The Deep Dive below builds the inverse when $\det M \neq 0$.

The inverse also tells how the answer moves with the data. Since $\mathbf{x} = M^{-1}\mathbf{b}$, column $j$ of $M^{-1}$ is the change in the mix when TAIEX's return on day $j$ rises by one point (Exhibit 8). Had TAIEX fallen 8.70% on 7 April, not 9.70%, the Semiconductor multiple would fall by 0.265 and the FinancialInsurance one rise by 0.468.

## Exhibit 8: The inverse of the 7–9 April matrix




Note: row $i$ is the multiple $x_i$ of Semiconductor, FinancialInsurance or Food ($i = 1, 2, 3$), and column $j$ the day (7, 8, 9 April). Entry $(i, j)$ is the change in $x_i$ per percentage point of TAIEX's return on day $j$. Computed exactly by Gauss–Jordan elimination and rounded to three decimals.

**EXAMPLE 6**

## LM7's Kelly fractions through the inverse

LM7 estimated, for 0050 and 00679B, an annualized covariance matrix with $\sigma_1^2 = 0.040425$, $\sigma_2^2 = 0.018214$ and $\sigma_{12} = -0.002866$, and excess returns $\boldsymbol{\mu} - r_f\mathbf{1} = (0.22751,\ -0.02547)^\top$ over an illustrative $r_f = 1.5\%$. Invert $\Sigma$ with (9.5) and compute the Kelly fractions $\Sigma^{-1}(\boldsymbol{\mu} - r_f\mathbf{1})$.

**SOLUTION**

The determinant is

$$\begin{aligned}\det\Sigma &= 0.040425(0.018214)\\ &\quad - 0.002866^2\\ &= 7.28087\times 10^{-4}.\end{aligned}$$

By (9.5),

$$\begin{aligned}\Sigma^{-1} &= \frac{1}{\det\Sigma}\begin{pmatrix}0.018214 & 0.002866\\ 0.002866 & 0.040425\end{pmatrix}\\ &= \begin{pmatrix}25.0162 & 3.9363\\ 3.9363 & 55.5222\end{pmatrix}.\end{aligned}$$

Multiplying by the excess returns gives the Kelly fractions

$$\begin{aligned}f_1^* &= 25.0162(0.22751)\\ &\quad - 3.9363(0.02547) = 5.5912,\\ f_2^* &= 3.9363(0.22751)\\ &\quad - 55.5222(0.02547) = -0.5186,\end{aligned}$$

LM7's $(5.59,\ -0.52)^\top$. The product $\Sigma\Sigma^{-1}$ is $I_2$ exactly; with the four-decimal entries printed here it differs from $I_2$ by at most 0.000002.

The inverse is a concept first. To solve one system, `np.linalg.solve` factors $M$ once and applies the elimination to $\mathbf{b}$ alone. `np.linalg.inv` applies it to all $n$ columns of $I_n$, so it costs more. Its $M^{-1}\mathbf{b}$ also has weaker error bounds.

Solving through the inverse is not backward stable: its answer need not be the exact solution of a slightly changed system. So the residual $M\mathbf{x} - \mathbf{b}$ can be larger, although $\mathbf{x}$ itself is usually about as accurate (Higham 2002a, ch. 14). Use `inv` for concepts such as $\Sigma^{-1}$; `solve` also takes many right-hand sides at once.

PYTHON

```python
import numpy as np

# rows 7, 8, 9 April; columns Semiconductor, FinancialInsurance, Food (%)
M = np.array([[-9.95, -9.89, -8.66],
              [-3.99,  0.53,  3.77],
              [-4.47, -5.61, -4.17]])
b = np.array([-9.70, -4.02, -5.79])     # TAIEX (%)
x = np.linalg.solve(M, b)               # elimination with partial pivoting
print(x.round(6))                       # [ 0.334105  1.446834 -0.916113]
print(round(np.linalg.det(M), 6))       # -71.591922
print(np.linalg.inv(M).round(3))        # the heatmap above
```

**導讀** **消去法與反矩陣**

高斯消去法（Gaussian elimination）用三種不改變解的列運算（row operation），把方程組化成上三角矩陣（upper triangular matrix），再從最後一列往上解回去，也就是後置法（back substitution）。每一步的對角元素是樞軸元（pivot），樞軸元相乘（加上換列的正負號）就是行列式（determinant）。NumPy 的 `np.linalg.solve` 做的正是這件事，而且每一步把絕對值最大的列換上來，以控制捨入誤差（rounding error）。反矩陣（inverse matrix）主要是概念工具：$\mathbf{x} = M^{-1}\mathbf{b}$ 說明解如何隨資料變動，LM7 的凱利比例就是 $\Sigma^{-1}(\boldsymbol{\mu} - r_f\mathbf{1})$。實際解題時先用 `solve`，不要先求反矩陣再相乘。

**KNOWLEDGE CHECK 5**

Why is `np.linalg.inv(M) @ b` the wrong habit for solving one system?

**ANSWER**

Computing `inv(M) @ b` does more work and gives up backward stability. `inv` applies elimination to the $n$ columns of $I_n$ and then multiplies, while `solve` applies it to $\mathbf{b}$ alone. The result can have a larger residual, although $\mathbf{x}$ itself is usually about as accurate (Higham 2002a, ch. 14).

**DEEP DIVE** **The adjugate formula $M^{-1} = \operatorname{adj} M / \det M$** optional · click to expand

Let $\operatorname{adj} M$ be the $n \times n$ matrix whose entry $(j, i)$ is the cofactor $(-1)^{i+j} \det M^{(ij)}$. Entry $(i, i)$ of $M \operatorname{adj} M$ is $\sum_j m_{ij}(-1)^{i+j} \det M^{(ij)}$, the cofactor expansion of $\det M$ along row $i$. For $k \neq i$, entry $(i, k)$ expands the matrix whose row $k$ is replaced by row $i$, which has two equal rows, so it is 0. Hence $M \operatorname{adj} M = (\det M)\, I_n$. Expanding along columns gives $(\operatorname{adj} M)M = (\det M)\, I_n$ in the same way, so when $\det M \neq 0$,

$$M^{-1} = \frac{\operatorname{adj} M}{\det M}.$$

For $n = 2$ this is (9.5). Entry $j$ of $\operatorname{adj} M\, \mathbf{b}$ expands $\det M_j(\mathbf{b})$ along column $j$, so $\mathbf{x} = M^{-1}\mathbf{b}$ is (9.4), which therefore solves the system (Strang 2016).

## 7 RANK AND THE NUMBER OF SOLUTIONS

- [ ] determine the rank, column space and null space of a matrix, and classify a system as having no, one or infinitely many solutions

**DEFINITION** **Independence, column space, null space and rank**

Vectors $\mathbf{m}_1, \ldots, \mathbf{m}_n$ are **linearly independent** if $x_1\mathbf{m}_1 + \cdots + x_n\mathbf{m}_n = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$. For an $m \times n$ matrix $M$, the **column space** $\operatorname{col} M$ is the set of all $M\mathbf{x}$. The **null space** $\operatorname{null} M$ is the set of all $\mathbf{x}$ with $M\mathbf{x} = \mathbf{0}$. The **rank** $\operatorname{rank} M$ is the largest number of independent columns. The **dimension** of a space of vectors is the largest number of independent vectors in it, so $\operatorname{rank} M = \dim \operatorname{col} M$.

Elimination works on any $m \times n$ matrix. When a column has no non-zero entry at or below the current row, it gets no pivot, and elimination moves to the next column. The result is an **echelon form**: each row starts further right than the row above, and zero rows come last. The pivot columns are independent and the other columns are combinations of them, so the rank is the number of pivots.

Each column without a pivot holds a **free variable**: give it any value, and back substitution finds the rest. With $\mathbf{b} = \mathbf{0}$, setting one free variable to 1 and the others to 0 gives a null-space vector $\mathbf{x}_h$. These vectors are independent and span $\operatorname{null} M$, one for each free column, so

$$\operatorname{rank} M + \dim \operatorname{null} M = n. \tag{9.7}$$

A system is solvable exactly when $\mathbf{b}$ lies in $\operatorname{col} M$. In echelon form, the entries of the transformed $\mathbf{b}$ beside zero rows of $M$ must then be 0: appending $\mathbf{b}$ adds no pivot. If $\mathbf{x}_p$ is one solution and $M\mathbf{x}_h = \mathbf{0}$, then $M(\mathbf{x}_p + \mathbf{x}_h) = \mathbf{b}$, and two solutions always differ by a null-space vector. Hence, for the system $M\mathbf{x} = \mathbf{b}$,

$$\begin{aligned} \text{solvable} &\iff \operatorname{rank} M = \operatorname{rank}[M \mid \mathbf{b}], \\ \text{solutions} &= \mathbf{x}_p + \operatorname{null} M. \end{aligned} \tag{9.8}$$

By (9.7), a solvable system has exactly one solution when $\operatorname{rank} M = n$, and infinitely many when $\operatorname{rank} M < n$.

**EXAMPLE 7**

## Two days, three sectors: a line of mixes

Find every mix of Semiconductor, FinancialInsurance and Food that reproduces TAIEX on 7 and 8 April.

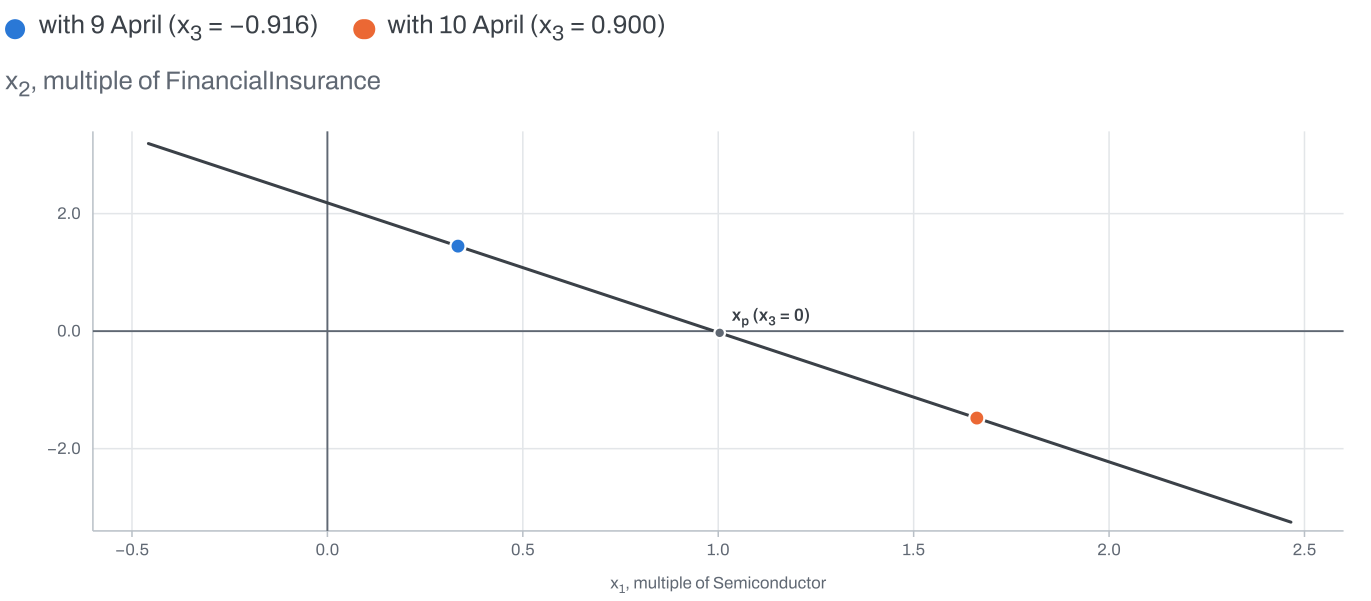
**SOLUTION**

Elimination on the first two rows of Exhibit 1 finds pivots in columns 1 and 2, so $\operatorname{rank} M = \operatorname{rank}[M \mid \mathbf{b}] = 2$ and $x_3$ is free. Back substitution leaves

$$\begin{aligned} x_1 &= 1.003671 + 0.730877\, x_3, \\ x_2 &= -0.028971 - 1.610943\, x_3. \end{aligned}$$

So $\mathbf{x} = \mathbf{x}_p + x_3\, \mathbf{x}_h$, with $\mathbf{x}_p = (1.003671,\ -0.028971,\ 0)^\top$ and $\mathbf{x}_h = (0.730877,\ -1.610943,\ 1)^\top$: a line of mixes, since $\dim \operatorname{null} M = 3 - 2 = 1$ by (9.7). Adding 9 April picks the point with $x_3 = -0.916113$, the mix of Example 4 (Exhibit 9).

**Exhibit 9: The 7–8 April line of mixes in the $(x_1, x_2)$ plane**




Note: $x_3$ runs from −2 to 2 along the line. A third day picks one point: 9 April gives $x_3 = -0.916113$ and 10 April gives $x_3 = 0.900269$.

EXAMPLE 8

### Four days, three sectors: no mix

Show that no mix reproduces TAIEX on all four days.

**SOLUTION**

The $4 \times 3$ matrix $M$ has rank 3, because its first three rows already give three pivots (Example 5). The $4 \times 4$ matrix $[M \mid \mathbf{b}]$ has determinant 65.603260, so its four columns are independent and its rank is 4. By (9.8) there is no solution: $\mathbf{b}$ lies outside $\operatorname{col} M$.

**Exhibit 10: Days used, ranks and solutions for the three-sector mix**

| Days | Equations × unknowns | $\operatorname{rank} M$ | $\operatorname{rank}[M \mid \mathbf{b}]$ | Solutions | Mix $(x_1, x_2, x_3)$ |
|---|---|---|---|---|---|
| 7–8 April | 2 × 3 | 2 | 2 | infinitely many | $\mathbf{x}_p + x_3\,\mathbf{x}_h$ |
| 7–9 April | 3 × 3 | 3 | 3 | exactly one | (0.334105, 1.446834, −0.916113) |
| 7–10 April | 4 × 3 | 3 | 4 | none | — |

Note: inputs from Exhibit 1; ranks in exact arithmetic. The line of the first row is Example 7's.

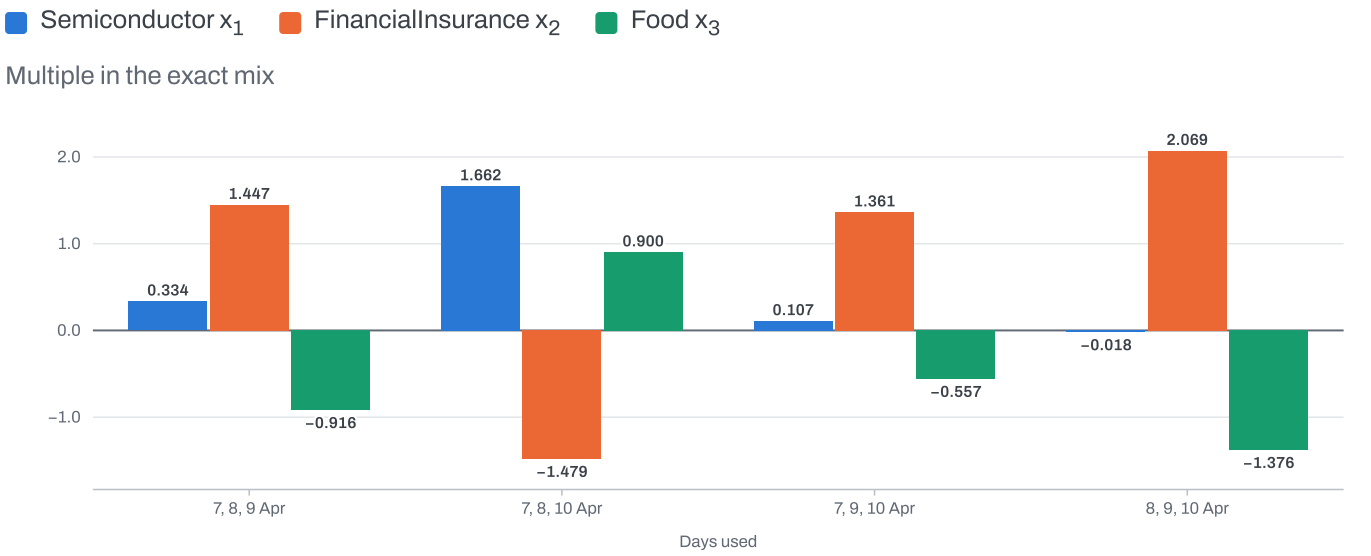
Exhibit 10 collects the three cases. With three days, every triple with a non-zero determinant fits exactly, so the fit itself carries no information. The four triples give four different mixes (Exhibit 11). FinancialInsurance's multiple runs from −1.479 (7, 8 and 10 April) to 2.069 (8, 9 and 10 April). And the 7–9 April mix, applied to the sectors' returns of 10 April, gives

$$\begin{aligned} &9.91(0.334105) + 8.36(1.446834) \\ &\quad + 5.72(-0.916113) = 10.17, \end{aligned}$$

against TAIEX's 9.25.

7 April adds less than it seems. It was a limit-down day for 0050 and TX (LM8), and the three sectors fell by 8.66% to 9.95%. Their row of returns lies 3.6° from $(-1, -1, -1)$, so the 7 April equation nearly fixes only the sum of the multiples.

## Exhibit 11: The exact mix on each triple of days




Note: the solution of the three-day system for each triple of 7–10 April, exact for the inputs of Exhibit 1. Determinants: −71.591922 (7, 8, 9), 22.568247 (7, 8, 10), −29.587887 (7, 9, 10) and 49.300524 (8, 9, 10).

> **NOTE** **Research hygiene**
>
> Each square fit is an **in-sample** exact fit: when $\det M \neq 0$ it always succeeds, so it carries no evidence about how TAIEX is built. Comparing the four triples is a **stability check**, and the mix fails it. The 10 April comparison is one illustration, not an **out-of-sample** test. Section 8's ranks, weights and correlation are **descriptive**: they summarize 2008–2026 and forecast nothing.
>
> With more days than sectors there is usually no exact solution; LM12 finds the multiples that miss by the least. Fitting over hundreds of days is a regression (LM23).

> **PITFALL** **One instrument twice: TX and MTX**
>
> Adding the Mini-TAIEX future (MTX) to TX adds no new direction. Both settle on TAIEX, and TX pays NT$200 per index point against MTX's NT$50 (TAIFEX contract specifications; LM2). Per point of the index, one TX contract moves exactly as four MTX contracts. For the same index-point moves, their columns of daily P&L per contract are proportional (200 against 50), so they have rank 1.
>
> A TX-plus-MTX hedge is a line of answers, $\mathbf{x}_p + x_2\,(-\frac{1}{4}, 1)^\top$, not a better hedge. This follows from the contract terms, not from settlement data (Problem 9).

**KNOWLEDGE CHECK 6**

Each augmented matrix below is in echelon form, for unknowns $x_1, x_2, x_3$. Does its system have no solution, one, or infinitely many?

$$\text{(a)}\ \left(\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\text{(b)}\ \left(\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 4 & 3 \end{array}\right)$$

$$\text{(c)}\ \left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 5 \end{array}\right)$$

**ANSWER**

(a) Infinitely many: both ranks are 2, and $x_2$ is free. (b) One: three pivots, so both ranks are 3. (c) None: the last row says $0 = 5$, so $\operatorname{rank}[M \mid \mathbf{b}] = 3$ exceeds $\operatorname{rank} M = 2$.

**DEEP DIVE** **Row rank equals column rank** optional · click to expand

The **row rank** is the largest number of independent rows. A row operation $EM$, with $E$ invertible, keeps all combinations of the rows and hence the row rank. It also keeps every relation among the columns, since $EM\mathbf{x} = \mathbf{0}$ exactly when $M\mathbf{x} = \mathbf{0}$, and hence the column rank. In echelon form both ranks equal the number of pivots (Strang 2016).

## 8 PERFECT COLLINEARITY IN RETURNS, AND PYTHON

- ☐ identify perfect collinearity in a matrix of returns, and explain why a daily-rebalanced mix creates it and a buy-and-hold mix does not
- ☐ solve and test linear systems in NumPy with `solve`, `inv`, `det` and `matrix_rank`, and interpret their floating-point output

Now let the rows be dates and the columns assets, as in the $T \times N$ return matrix $X$ of LM8. Its columns are dependent exactly when some fixed non-zero combination of the columns is 0 on every date, a case called **perfect collinearity**. By (9.7),

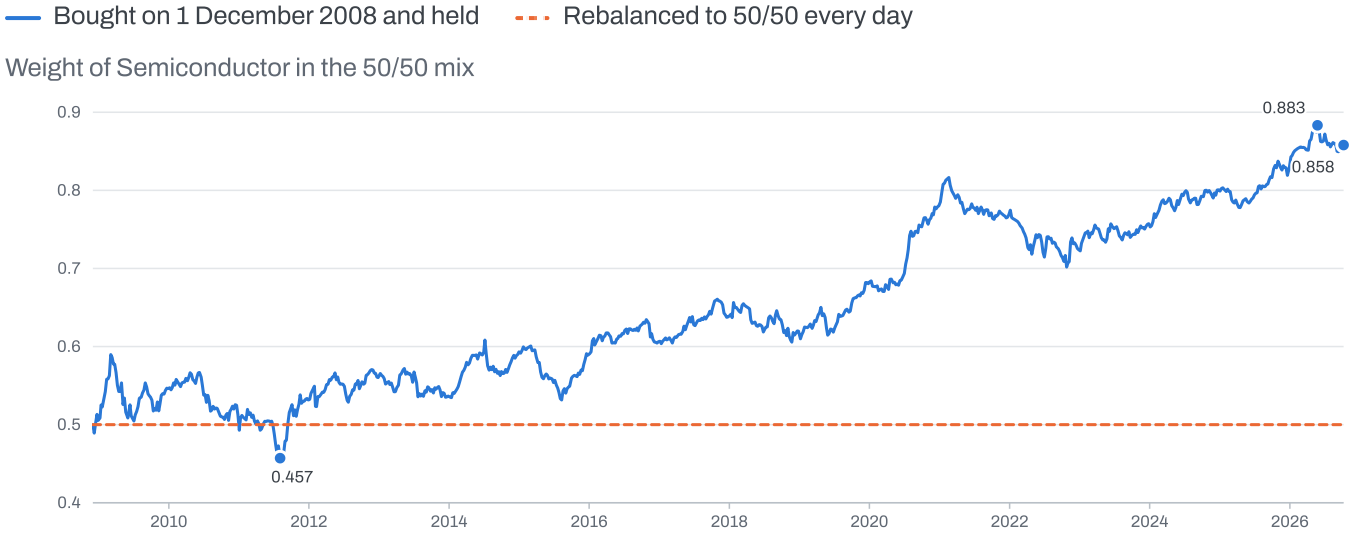
$$\operatorname{rank} X < N \iff X\mathbf{x} = \mathbf{0} \ \text{ for some } \mathbf{x} \neq \mathbf{0}. \tag{9.9}$$

Take the daily returns of Semiconductor and FinancialInsurance on the Volume II sector sample: $T =$ 4,382 returns from 2 December 2008 to 8 October 2026. Add a 50/50 mix of the two as a third column. A

mix **rebalanced daily** to 50/50 returns $\frac{1}{2}R_{S,t} + \frac{1}{2}R_{F,t}$ on every date. So $\mathbf{x} = (-\frac{1}{2}, -\frac{1}{2}, 1)^\top$ solves $X\mathbf{x} = \mathbf{0}$, and the rank is 2.

A mix **bought and held** from the close of 1 December 2008 lets its weights drift with prices. Its return on date $t$ is $w_{t-1}R_{S,t} + (1 - w_{t-1})R_{F,t}$, where $w_{t-1}$ is Semiconductor's weight at the previous close. That weight rose from 0.50 to 0.858 by 8 October 2026 (Exhibit 12). Along the way it fell to 0.457 on 4 August 2011 and peaked at 0.883 on 25 May 2026. No fixed $\mathbf{x}$ matches a changing weight, so the rank is 3.

**Exhibit 12: Semiconductor's weight in two 50/50 mixes with FinancialInsurance, December 2008 – October 2026**




Source: FinMind `TaiwanStockPrice`, closes of the sector price indices, retrieved 11 October 2026, with the copied closes that `data/vol2_series.py` repairs from the 13:30:00 values of `TaiwanStockEvery5SecondsIndex`. Both mixes start at 50/50 at the close of 1 December 2008; the rebalanced mix returns to 50/50 at every close. Computations: `data/lm09_charts.py`.

Real indices are nearly collinear, not exactly. TWSE's Electronic index covers the stocks of its eight electronics sub-industries (TWSE 2007). Like every sector index it weights each stock by its market value (TWSE 2023), so its weights drift with prices (LM27). The 4,382 × 9 matrix of Electronic and its eight sub-industries therefore has full rank 9, although Electronic's correlation with Semiconductor alone is 0.952591.

LM11 measures how close such columns come to dependence with the condition number, and LM12 decides when to treat them as dependent. Volume II's sector set U28 holds the 28 leaf indices among the 30 TWSE sector indices that FinMind serves daily. It excludes the aggregates Electronic and ChemicalBiotechnologyMedicalCare, and its return matrix has rank 28.

EXAMPLE 9

## A mix column on 7–9 April

Replace Food in Example 3 by the daily-rebalanced 50/50 mix of Semiconductor and FinancialInsurance, $\mathbf{m}_3 = (\mathbf{m}_1 + \mathbf{m}_2)/2$, which is $(-9.92,\ -1.73,\ -5.04)^\top$. Find the rank, the null space and the solutions.

**SOLUTION**

The third column is $\frac{1}{2}\mathbf{m}_1 + \frac{1}{2}\mathbf{m}_2$, so $M\mathbf{x}_h = \mathbf{0}$ for $\mathbf{x}_h = (-\frac{1}{2}, -\frac{1}{2}, 1)^\top$ and $\det M = 0$. Elimination finds two pivots, so rank $M = 2$ and null $M$ is the line through $\mathbf{x}_h$. On 8 and 9 April only Example 1's combination of $\mathbf{m}_1$ and $\mathbf{m}_2$ matches $\mathbf{b}$, and it gives −12.35% on 7 April, not −9.70%. So rank$[M \mid \mathbf{b}] = 3$, and by (9.8) there is no solution.

Floating point keeps about 16 significant digits (LM2), so an exact 0 rarely survives. For the matrix of Example 9, `np.linalg.det` may print about $10^{-14}$ rather than 0. `solve` may return entries of order $10^{15}$ along $\mathbf{x}_h$ without an error, because LAPACK stops only at an exactly zero pivot. `matrix_rank` applies a tolerance (LM12) and reports 2. So test singularity with the rank, never with `det(M) == 0`.

PYTHON

```python
Mmix = np.column_stack([M[:, 0], M[:, 1], (M[:, 0] + M[:, 1]) / 2])
print(np.linalg.matrix_rank(Mmix))        # 2: a non-zero combination is 0
print(abs(np.linalg.det(Mmix)) < 1e-10)   # True: det is tiny but may not be 0.0
four = np.vstack([M, [9.91, 8.36, 5.72]])           # add 10 April
Mb = np.column_stack([four, [-9.70, -4.02, -5.79, 9.25]])
print(np.linalg.matrix_rank(four), np.linalg.matrix_rank(Mb))   # 3 4
# np.linalg.solve(four, ...) raises LinAlgError:
#   Last 2 dimensions of the array must be square
```

**IN PYTHON** **Companion lab**

The Polars-first notebook `LM09_lab.ipynb` rebuilds the case, the exhibits and the computational examples from the sector closes. It also solves the computational practice problems.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

**導讀** **剛好擬合不代表有意義；完全共線**

三天、三個產業、三個未知數：只要行列式（determinant）不為 0，就一定解得出一組剛好重現 TAIEX 報酬的倍數，所以剛好擬合本身不提供任何證據。換一組三天，金融保險類的倍數可以從 −1.479 跳到 2.069；用 7–9 日的解去算 10 日，得到 +10.17%，實際是 +9.25%。報酬矩陣的秩（rank）小於行數，等於說各行某個係數不全為 0 的固定組合每天都是 0，這就是完全共線性（perfect collinearity）。每天再平衡的 50/50 組合正是如此，秩只有 2；買進持有的組合權重隨價格漂移，從 0.50 漂到 0.858，秩是 3。

**KNOWLEDGE CHECK 7**

Beside the returns of Semiconductor and FinancialInsurance, which of these third columns make the rank lower than 3 on every sample?

- a 70/30 mix rebalanced daily
- the spread $R_{S,t} - R_{F,t}$
- a 70/30 mix bought and held

**ANSWER**

The rebalanced mix and the spread do. With either as the third column, $X\mathbf{x} = \mathbf{0}$ on every date, for $\mathbf{x} = (-0.7, -0.3, 1)^\top$ and $\mathbf{x} = (-1, 1, 1)^\top$ respectively. The held mix has drifting weights (Exhibit 12), so on a long sample no fixed non-zero combination is zero on every date.

## SUMMARY

- A linear system $M\mathbf{x} = \mathbf{b}$ reads by rows, as lines or planes that meet, and by columns (9.1), as a mix that reaches $\mathbf{b}$. On 8 and 9 April, 1.035 of Semiconductor and 0.207 of FinancialInsurance reproduce TAIEX.
- The determinant (9.2) is the signed area or volume of the columns and is multiplicative (9.3). It is 0 exactly when the columns are dependent; judge a small value against their lengths.
- Cramer's rule (9.4) solved the 7–9 April system, $\mathbf{x} = (0.334105,\ 1.446834,\ -0.916113)^\top$, with $(n+1)!$ terms against about $\frac{2}{3}n^3$ operations for elimination.
- Elimination with back substitution and partial pivoting is what `np.linalg.solve` runs. Its pivots multiply to $\pm \det M$, here −71.591922.
- The inverse (9.5) exists exactly under (9.6). It explains answers, such as LM7's $\Sigma^{-1}(\boldsymbol{\mu} - r_f\mathbf{1}) = (5.59,\ -0.52)^\top$, while `solve` computes them.
- Rank, (9.7) and (9.8) separate no, one and infinitely many solutions: four, three and two days of the case.
- An exact fit on three days carries no evidence. FinancialInsurance's multiple ran from −1.479 to 2.069 across the four triples, and the 7–9 April mix gave +10.17% for 10 April, against +9.25%.
- A return matrix loses rank exactly when a fixed non-zero combination of its columns is 0 on every date (9.9). A column holding a daily-rebalanced mix creates one; a buy-and-hold mix does not. Floating point can turn the zero determinant of Example 9 into about $10^{-14}$, so test with `matrix_rank`.

## PRACTICE PROBLEMS

1. You swap the rows of 7 and 8 April in the matrix of Example 3 and double the row of 9 April. The new determinant equals $\det M$ times:

   **A.** $2$

   **B.** $-1$

   **C.** $-2$

2. A system of four equations in three unknowns has $\operatorname{rank}[M \mid \mathbf{b}] = 4$. How many solutions does it have?

   **A.** none

   **B.** exactly one

   **C.** infinitely many

3. A matrix holds three columns of daily returns over a long sample: Semiconductor, FinancialInsurance and a 50/50 mix of the two. Which mix makes its rank lower than 3 on every sample?

   **A.** a mix bought at 50/50 on the first day and held

   **B.** a mix rebalanced to 50/50 at every close

   **C.** either, because both start at 50/50

4. Find the mix of Semiconductor and Food that reproduces TAIEX on 7 and 10 April, once by Cramer's rule (9.4) and once by (9.5).

5. Solve the system of 8, 9 and 10 April, with the three sectors, by elimination with partial pivoting. Report the row swaps, the pivots and $\mathbf{x}$, and compare $\mathbf{x}$ with Example 5.

6. Compute the determinant of the 7, 8 and 10 April matrix by cofactor expansion along the first row. Check it with the pivots of elimination without swaps.

7. For a constant $a$, consider the system with augmented matrix

$$[M \mid \mathbf{b}] = \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & 7 & 3 \\ 1 & 2 & 4 & a \end{array}\right).$$

(a) Find $\operatorname{rank} M$ and describe $\operatorname{col} M$. (b) For $a$

which is $a$ the system solvable? (c) For that $a$, give $\mathbf{x}_h$ and every solution.

8. Three illustrative bonds with face value 100 pay annual coupons: 2% for one year, 3% for two years and 4% for three years. They cost 100.50, 102.55 and 106.25. Write the system for the discount factors $v_t = (1+z_t)^{-t}$ of LM6's spot curve, here $v_1, v_2, v_3$. Explain why it needs no elimination, solve it, and give the spot rates $z_t$.

9. A trader holds $x_1$ TX and $x_2$ MTX contracts. On two illustrative days TAIEX moves −300 and +150 points. (a) Write the $2 \times 2$ system for a target P&L $\mathbf{b}$ in NT$, and show that its matrix is singular. (b) Which targets can be reached? (c) Find $\mathbf{x}_h$ and interpret it.

10. *(Python)* For the system of Example 3, compare `np.linalg.solve(M, b)` with `np.linalg.inv(M) @ b`. Then try `solve` on all four days, and compute `matrix_rank` of $M$ and of $[M \mid \mathbf{b}]$ for the four days.

11. *(Python)* Use Section 8's sample: closes from 1 December 2008 to 8 October 2026, repaired as in the lab, and 4,382 returns. Build the daily-rebalanced and the buy-and-hold 50/50 mixes of Semiconductor and FinancialInsurance. Compute `matrix_rank` of each three-column matrix, the final buy-and-hold weight in Semiconductor and the largest absolute daily gap between the two mixes' returns, in percentage points.

## SOLUTIONS

1. **C** is correct. The swap changes the sign (rule 1), and doubling a row doubles the determinant (rule 2), so the new determinant is $-2 \det M = 143.183844$. A ignores the swap, and B ignores the doubling.

2. **A** is correct. The matrix $M$ has only three columns, so $\operatorname{rank} M \le 3 < 4$, and by (9.8) there is no solution. B needs both ranks equal to 3, and C needs them equal and below 3.

3. **B** is correct. The rebalanced mix returns the average of the two sectors' returns on every date, so $(-\tfrac{1}{2}, -\tfrac{1}{2}, 1)^{\top}$ lies in the null space (9.9) and the rank is below 3. The held mix's weights drift (Section

8), so on a long sample no fixed non-zero combination is zero and the rank is 3. C fails for the same reason.

4. The determinant and, by (9.4), the mix are

$$\begin{aligned}\det M &= (-9.95)(5.72) - (-8.66)(9.91)\\ &= -56.914 + 85.8206 = 28.9066,\\ x_1 &= \frac{(-9.70)(5.72) - (-8.66)(9.25)}{28.9066}\\ &= \frac{24.621}{28.9066} = 0.85174,\\ x_2 &= \frac{(-9.95)(9.25) - (-9.70)(9.91)}{28.9066}\\ &= \frac{4.0895}{28.9066} = 0.14147.\end{aligned}$$

   By (9.5) the inverse is

$$M^{-1} = \frac{1}{28.9066}\begin{pmatrix} 5.72 & 8.66 \\ -9.91 & -9.95 \end{pmatrix}.$$

   Multiplying it by $\mathbf{b}$ gives the same mix. Its multiples nearly sum to 1 (0.993), while on 7 and 9 April the same two sectors needed −3.48 and 5.12 (Example 2).

5. Column 1's largest entry in absolute value is 9.91, so the 10 April row moves to the top. The multipliers are $\ell_{21} = -0.451060$ (9 April) and $\ell_{31} = -0.402624$ (8 April). In column 2 the 8 April row now has the larger entry, 3.895933 against −1.839142, so it moves up, and $\ell_{32} = -0.472067$.

   The pivots are 9.91, 3.895933 and 1.276928, and back substitution gives $\mathbf{x} = (-0.017926,\ 2.069318,\ -1.376197)^{\top}$. Two swaps keep the sign, so the exact pivots multiply to $\det M = 49.300524$. Against Example 5, changing one day moves FinancialInsurance's multiple from 1.447 to 2.069.

6. The minors of the 7 April row are $\det M^{(11)} = -28.4856$, $\det M^{(12)} = -60.1835$ and $\det M^{(13)} = -38.6087$, so

$$\begin{aligned}\det M &= -9.95(-28.4856)\\ &\quad + 9.89(-60.1835)\\ &\quad - 8.66(-38.6087)\\ &= 22.568247,\end{aligned}$$

   the sum of 283.43172, −595.214815 and 334.351342. Without swaps, elimination keeps the first two rows of Example 5, so $d_1 = -9.95$ and $d_2 = 4.495940$. After step 1 the 10 April row is $(0,\ -1.490241,\ -2.905186)$; then, in exact arithmetic, $\ell_{32} = -0.331464$ and $d_3 = -0.504492$. The exact pivots multiply to 22.568247.

7. (a) Subtract twice row 1 from row 2 and row 1 from row 3, then row 2 from row 3:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & a-2 \end{array}\right).$$

The pivots are in columns 1 and 3, so rank $M = 2$. The column space $\operatorname{col} M$ is the plane of all combinations of $(1,2,1)^{\top}$ and $(3,7,4)^{\top}$; the second column is twice the first. (b) The last row reads $0 = a - 2$, so the system is solvable only for $a = 2$. (c) With $x_2$ free, $\mathbf{x}_h = (-2,1,0)^{\top}$. The other rows give $x_3 = 1$ and $x_1 = -2 - 2x_2$, so the solutions are $(-2,0,1)^{\top} + x_2\,(-2,1,0)^{\top}$, and $2 + 1 = 3$ as (9.7) requires.

8. Each bond pays its coupon every year and 100 more at maturity, so the matrix is lower triangular:

$$\begin{pmatrix} 102 & 0 & 0 \\ 3 & 103 & 0 \\ 4 & 4 & 104 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 100.50 \\ 102.55 \\ 106.25 \end{pmatrix}.$$

No elimination is needed. Forward substitution solves the rows from the top:

$$\begin{aligned} v_1 &= \frac{100.50}{102} = 0.985294, \\ v_2 &= \frac{102.55 - 3v_1}{103} = 0.966933, \\ v_3 &= \frac{106.25 - 4v_1 - 4v_2}{104} = 0.946549. \end{aligned}$$

The implied annual-compounding spot rates $z_t = v_t^{-1/t} - 1$ are 1.4925%, 1.6955% and 1.8480%. They rise with maturity like LM6's illustrative curve.

9. (a) Each contract's daily P&L is its multiplier times the index move, so

$$\begin{pmatrix} -60{,}000 & -15{,}000 \\ 30{,}000 & 7{,}500 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \mathbf{b},$$

whose matrix has determinant $-450{,}000{,}000 + 450{,}000{,}000 = 0$. (b) Both columns are multiples of $(-300, 150)^{\top}$, so only targets proportional to it can be reached: $b_1 = -2b_2$. (c) The null vector is $\mathbf{x}_h = (-\tfrac{1}{4}, 1)^{\top}$: short a quarter of a TX contract against one MTX, and the P&L is 0 on every day. So every reachable target has a line of solutions.

10. The two answers agree to about 16 significant digits; their largest difference, of order $10^{-16}$, varies from machine to machine. On the $4 \times 3$ matrix, `solve` raises `LinAlgError` with the message "Last 2 dimensions of the array must be square". `matrix_rank` gives 3 for $M$ and 4 for $[M \mid \mathbf{b}]$, so there is no exact solution, as in Example 8.

11. The ranks are 2 for the rebalanced mix and 3 for the buy-and-hold mix. The buy-and-hold weight in Semiconductor ends at 0.858 on 8 October 2026. The largest absolute daily gap between the two mixes' returns is 2.57 percentage points.

## GLOSSARY

| Term | 中文 | Meaning |
|---|---|---|
| **Linear system** | 線性方程組 | $M\mathbf{x} = \mathbf{b}$: $m$ linear equations in $n$ unknowns. |
| **Augmented matrix** | 增廣矩陣 | $[M \mid \mathbf{b}]$, the matrix with $\mathbf{b}$ appended as a last column. |
| **Row picture, column picture** | 按列讀、按行讀 | Lines or planes that meet; columns combined to reach $\mathbf{b}$, (9.1). |
| **Replicating mix** | 複製組合 | Multiples of assets whose returns reproduce a target's; not weights. |
| **Determinant** | 行列式 | Signed area or volume spanned by the columns, (9.2), (9.3). |
| **Cofactor** | 餘因子 | $(-1)^{i+j} \det M^{(ij)}$; the cofactor expansion sums entries times cofactors. |
| **Cramer's rule** | 克拉瑪法則 | $x_j = \det M_j(\mathbf{b}) / \det M$, (9.4). |
| **Gaussian elimination** | 高斯消去法 | Row operations to triangular form, then back substitution. |
| **Pivot** | 樞軸元 | The diagonal entry that clears the entries below it. |
| **Back substitution** | 後置法 | Solving a triangular system from the last equation up. |
| **Partial pivoting** | 部分軸元消元法 | Moving the row with the largest entry in the pivot column into the pivot position. |
| **Inverse matrix** | 反矩陣 | $M^{-1}$ with $MM^{-1} = M^{-1}M = I_n$, (9.5), (9.6). |
| **Singular matrix** | 奇異矩陣 | A square matrix with $\det M = 0$; it has no inverse. |
| **Gauss–Jordan elimination** | 高斯-約旦消去法 | Row reduction of $[M \mid I_n]$ to $[I_n \mid M^{-1}]$. |
| **Linear independence** | 線性獨立 | No combination of the vectors with a non-zero coefficient is $\mathbf{0}$. |
| **Column space** | 行空間 | All combinations $M\mathbf{x}$ of the columns. |
| **Null space** | 零空間 | All $\mathbf{x}$ with $M\mathbf{x} = \mathbf{0}$. |
| **Rank** | 秩 | Number of independent columns, or of pivots, (9.7). |
| **Free variable** | 自由變數 | An unknown whose column has no pivot. |
| **Particular solution** | 特解 | One solution $\mathbf{x}_p$ of $M\mathbf{x} = \mathbf{b}$, (9.8). |

| Term | 中文 | Meaning |
|---|---|---|
| **Perfect collinearity** | 完全共線性 | A fixed non-zero combination of the columns that is 0 on every row, (9.9). |

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